Pharmacovigilance Intelligence Platform:

From Raw Adverse Event Reports to AI-Powered Safety Signals

Submitted by
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Technology Stack:
AWS S3
Dataiku DSS
Snowflake
Snowflake Cortex ML
Power BI

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INTRODUCTION

1.1 Purpose

This Test Report documents the verification and validation activities performed for the Pharmacovigilance Intelligence Platform: From Raw Adverse Event Reports to AI-Powered Safety Signals. The objective of testing was to validate the functionality, data accuracy, and reliability of the end-to-end data pipeline, including data ingestion, transformation, analytical modeling, AI/ML forecasting, and reporting components. The testing covered AWS S3, Dataiku DSS, Snowflake Data Cloud, Snowflake Cortex Forecast, and Power BI to ensure correct data flow, transformation accuracy, analytical output validation, and dashboard reliability.

1.2 Project Overview

The Pharmacovigilance Intelligence Platform is a cloud-based data engineering solution designed to transform raw adverse event reports into structured analytical insights for safety monitoring and predictive analysis.

The platform implements an end-to-end data pipeline consisting of:
Data ingestion from AWS S3.
Data cleansing and validation using Dataiku DSS.
Analytical data modeling and processing in Snowflake.
Adverse event trend forecasting using Snowflake Cortex Forecast.
Business intelligence reporting through Power BI.

The testing process focused on validating data quality, transformation logic, analytical datasets, forecasting inputs, and reporting outputs.

1.3Scope of Testing

The testing activities covered the following areas:

Data Pipeline Validation

Validation of raw data ingestion into Snowflake.
Verification of data movement across pipeline layers.

Data Quality Validation

Validation of missing value handling.
Duplicate record identification and removal.
Data type and format validation.
Standardization of inconsistent values.

Data Warehouse Validation

Validation of Snowflake tables, views, and dimensional model.
Verification of analytical data accuracy.

AI/ML Validation

Validation of forecasting dataset preparation.

Verification of Snowflake Cortex Forecast configuration and output.

Reporting Validation

Validation of Power BI dashboard metrics and visualizations.

Out of Scope:

Production deployment testing.

User acceptance testing.

Regulatory compliance validation.

Test Environment

2.1 Environment Overview

Testing was performed in a cloud-based environment using AWS, Dataiku DSS, Snowflake, and Power BI. The environment was configured to validate the complete data engineering pipeline, from raw data ingestion through analytical reporting and AI/ML forecasting.

Component	Purpose
AWS S3	Raw data storage
Dataiku DSS	Data profiling, cleansing, and transformation
Snowflake	Cloud data warehouse and analytical processing
Snowflake Cortex Forecast	AI/ML forecasting
Power BI	Dashboard visualization and reporting

2.2 Technology Stack

Technology	Version/Configuration	Purpose
AWS S3	Cloud Storage	Raw dataset storage
Dataiku DSS	Community/Academic Edition	Data preparation and quality validation
Snowflake	Enterprise Data Cloud	Data warehousing
Snowflake Cortex Forecast	Built-in ML	Time-series forecasting
Power BI Desktop	Latest Version	Reporting and visualization

2.3 Snowflake Configuration

Parameter	Value
Warehouse	PHARMA_WH
Database	PHARMA_DB
Schemas	RAW,STAGING,CURATED
Warehouse Size	X-Small
Auto Suspend	60 Seconds
Auto Resume	Enabled

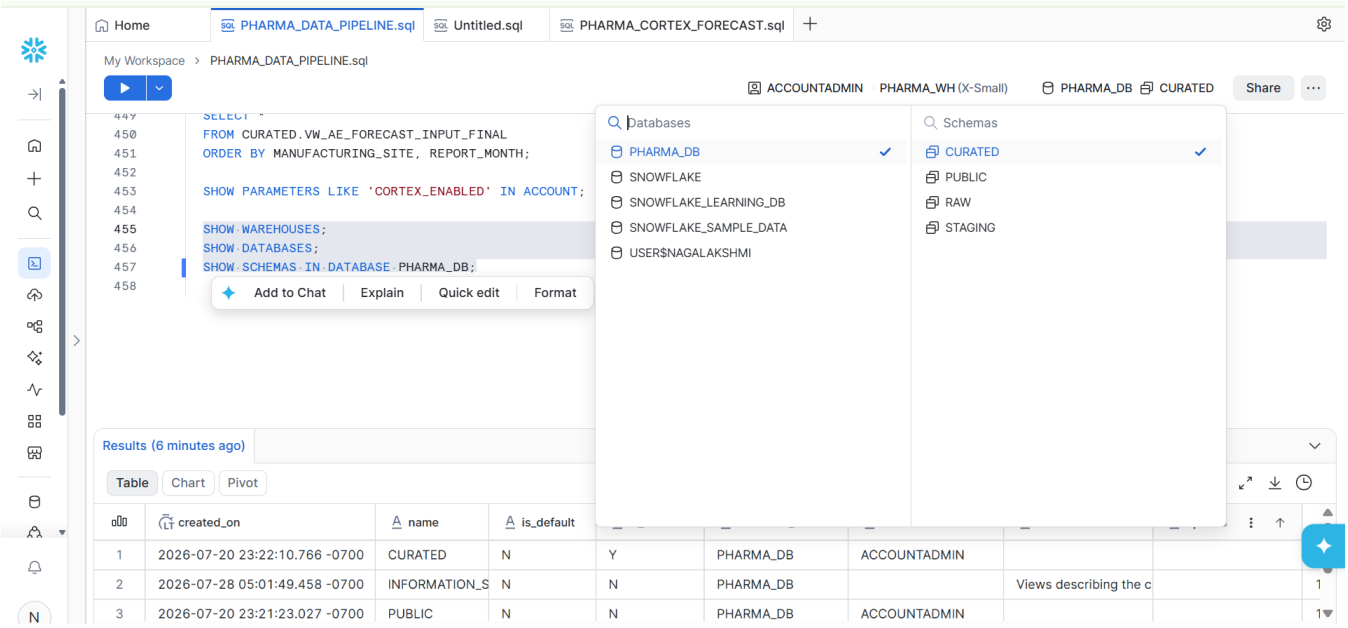
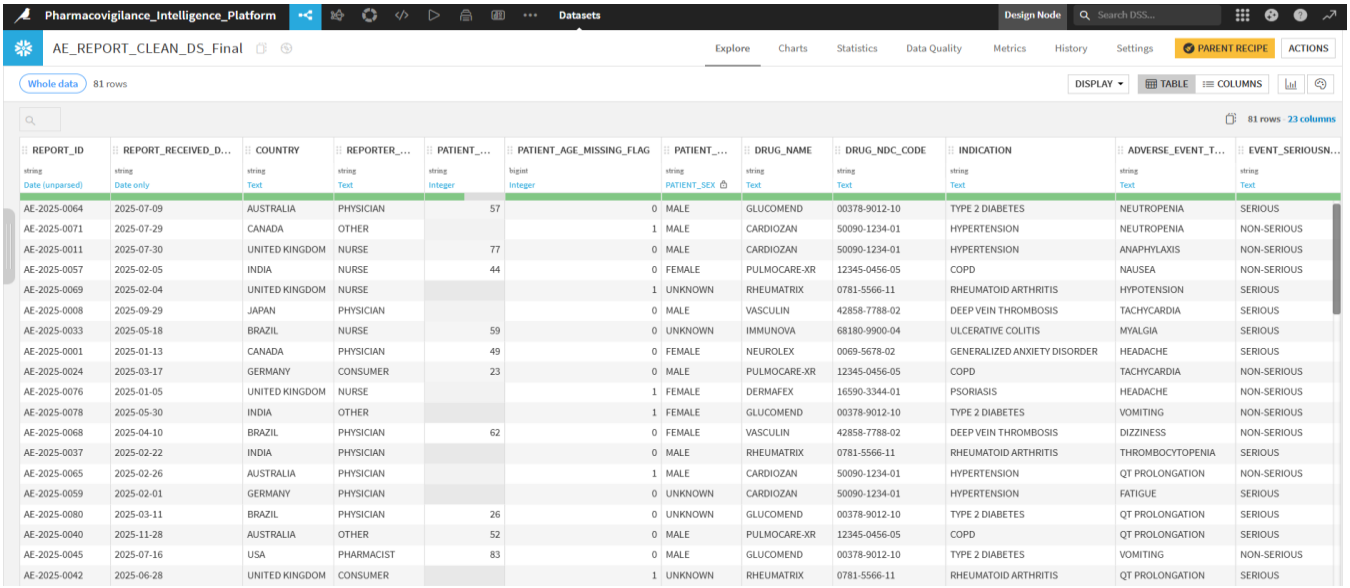


Figure 2.1: Snowflake Test Environment Configuration

2.4 Test Dataset

Attribute	Value
Dataset Name	Adverse Event Reports
Source Format	CSV
Source Storage	AWS S3
Initial Records	83
Records after Data Cleaning	81
Number of Columns	23



REPORT_ID	REPORT_RECEIVED_D...	COUNTRY	REPORTER_...	PATIENT_...	PATIENT_AGE_MISSING_FLAG	PATIENT_...	DRUG_NAME	DRUG_NDC_CODE	INDICATION	ADVERSE_EVENT_T...	EVENT_SERIOUSN...
string	string	string	string	string	bigint	string	string	string	string	string	string
Date (unparsed)	Date only	Text	Text	Integer	Integer	PATIENT_SEX	Text	Text	Text	Text	Text
AE-2025-0064	2025-07-09	AUSTRALIA	PHYSICIAN	57	0	MALE	GLUCOMEND	00378-9012-10	TYPE 2 DIABETES	NEUTROPENIA	SERIOUS
AE-2025-0071	2025-07-29	CANADA	OTHER		1	MALE	CARDIOZAN	50090-1234-01	HYPERTENSION	NEUTROPENIA	NON-SERIOUS
AE-2025-0011	2025-07-30	UNITED KINGDOM	NURSE	77	0	MALE	CARDIOZAN	50090-1234-01	HYPERTENSION	ANAPHYLAXIS	NON-SERIOUS
AE-2025-0057	2025-02-05	INDIA	NURSE	44	0	FEMALE	PULMOCARE-XR	12345-0456-05	COPD	NAUSEA	NON-SERIOUS
AE-2025-0069	2025-02-04	UNITED KINGDOM	NURSE		1	UNKNOWN	RHEUMATRIX	0781-5566-11	RHEUMATOID ARTHRITIS	HYPOTENSION	SERIOUS
AE-2025-0008	2025-09-29	JAPAN	PHYSICIAN		0	MALE	VASCULIN	42858-7788-02	DEEP VEIN THROMBOSIS	TACHYCARDIA	SERIOUS
AE-2025-0033	2025-05-18	BRAZIL	NURSE	59	0	UNKNOWN	IMMUNOVA	68180-9900-04	ULCERATIVE COLITIS	MYALGIA	SERIOUS
AE-2025-0001	2025-01-13	CANADA	PHYSICIAN	49	0	FEMALE	NEUROLEX	0069-5678-02	GENERALIZED ANXIETY DISORDER	HEADACHE	SERIOUS
AE-2025-0024	2025-03-17	GERMANY	CONSUMER	23	0	MALE	PULMOCARE-XR	12345-0456-05	COPD	TACHYCARDIA	NON-SERIOUS
AE-2025-0076	2025-01-05	UNITED KINGDOM	NURSE		1	FEMALE	DERMAFEX	16590-3344-01	PSORIASIS	HEADACHE	NON-SERIOUS
AE-2025-0078	2025-05-30	INDIA	OTHER		1	FEMALE	GLUCOMEND	00378-9012-10	TYPE 2 DIABETES	VOMITING	NON-SERIOUS
AE-2025-0068	2025-04-10	BRAZIL	PHYSICIAN	62	0	FEMALE	VASCULIN	42858-7788-02	DEEP VEIN THROMBOSIS	DIZZINESS	NON-SERIOUS
AE-2025-0037	2025-02-22	INDIA	PHYSICIAN		0	MALE	RHEUMATRIX	0781-5566-11	RHEUMATOID ARTHRITIS	THROMBOCYTOPENIA	SERIOUS
AE-2025-0065	2025-02-26	AUSTRALIA	PHYSICIAN		1	MALE	CARDIOZAN	50090-1234-01	HYPERTENSION	QT PROLONGATION	NON-SERIOUS
AE-2025-0059	2025-02-01	GERMANY	PHYSICIAN		0	UNKNOWN	CARDIOZAN	50090-1234-01	HYPERTENSION	FATIGUE	SERIOUS
AE-2025-0080	2025-03-11	BRAZIL	PHYSICIAN	26	0	UNKNOWN	GLUCOMEND	00378-9012-10	TYPE 2 DIABETES	QT PROLONGATION	SERIOUS
AE-2025-0040	2025-11-28	AUSTRALIA	OTHER	52	0	MALE	PULMOCARE-XR	12345-0456-05	COPD	QT PROLONGATION	SERIOUS
AE-2025-0045	2025-07-16	USA	PHARMACIST	83	0	MALE	GLUCOMEND	00378-9012-10	TYPE 2 DIABETES	VOMITING	NON-SERIOUS
AE-2025-0042	2025-06-28	UNITED KINGDOM	CONSUMER		1	UNKNOWN	RHEUMATRIX	0781-5566-11	RHEUMATOID ARTHRITIS	QT PROLONGATION	SERIOUS

Figure 2.2: Final Cleaned Dataset Used for Testing

2.5 Architecture

The test environment follows a cloud-native data engineering architecture in which raw adverse event data is ingested from AWS S3, processed using Dataiku DSS, stored and analyzed in Snowflake, forecasted using Snowflake Cortex Forecast, and visualized through Power BI dashboards.

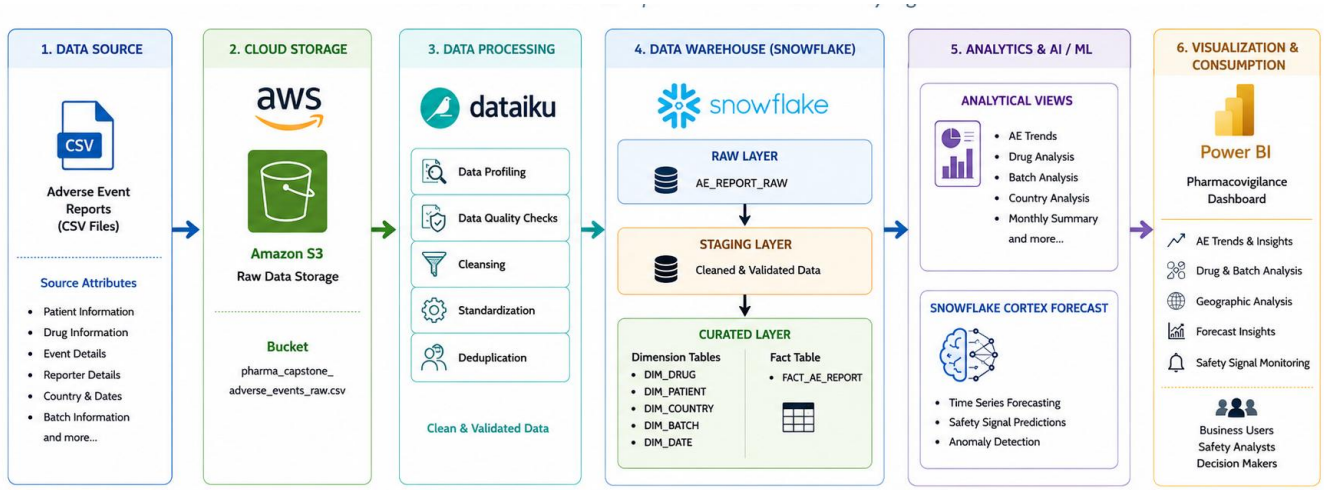


Figure 2.5: illustrates the end-to-end architecture of the implemented solution.

3. Test Summary

The testing activities were conducted to validate the end-to-end functionality of the Pharmacovigilance Intelligence Platform, ensuring that each component of the data pipeline met the defined functional and data quality requirements.

The following areas were included in the test execution:

Test Area	Objective	Status
Data Ingestion	Validate successful loading of raw data from AWS S3 into Snowflake	Passed
Data Quality	Verify data cleansing, standardization, and duplicate removal	Passed
Data Transformation	Validate Dataiku ETL workflows and transformation logic	Passed
Data Warehouse	Verify star schema, tables, views, and data integrity	Passed
AI/ML Forecasting	Validate Snowflake Cortex Forecast model execution and prediction output	Passed
Dashboard Validation	Verify Power BI visualizations and reported metrics	Passed

Overall Test Execution

Metric	Result
Total Test Areas	6
Passed	6
Failed	0
Blocked	0
Overall Status	**PASS**

4. Test Cases

4.1 Data Ingestion Testing

TC ID	Module	Test Scenario	Expected Result	Actual Result	Status
TC-001	AWS S3 Ingestion	Verify raw adverse event CSV file is available in the configured AWS S3 location.	Source file is accessible without corruption.	CSV file successfully accessed and used for ingestion.	Pass
TC-002	Snowflake RAW Load	Verify raw dataset is loaded into the RAW schema.	All source records are loaded successfully.	Raw dataset loaded successfully into Snowflake RAW schema.	Pass
TC-003	Data Validation	Verify source and target record counts match after ingestion.	Record count in Snowflake matches the source dataset.	Source and target record counts matched successfully.	Pass

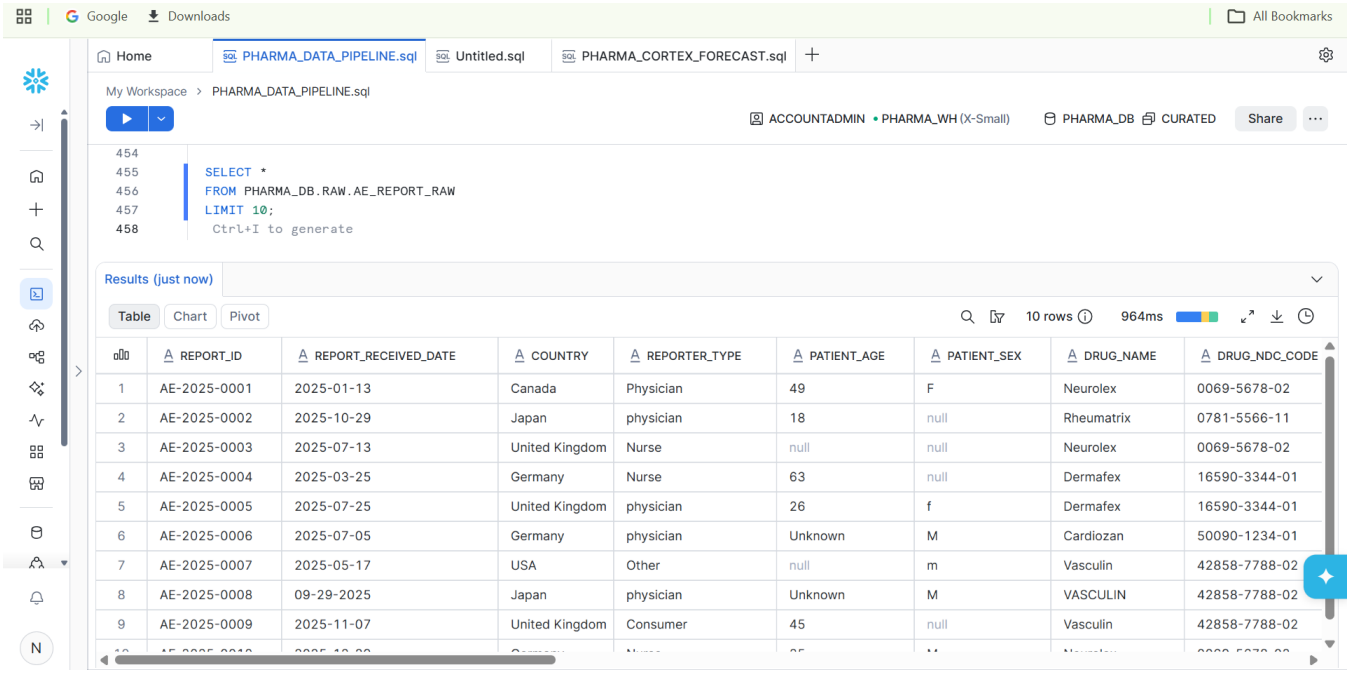


Figure 4.1: Raw Adverse Event Data Loaded into Snowflake RAW Schema

4.2 Data Quality Testing

TC ID	Module	Test Scenario	Expected Result	Actual Result	Status
TC-004	Duplicate Validation	Verify duplicate `REPORT_ID` records are identified and removed.	No duplicate `REPORT_ID` values remain after cleansing.	Duplicate records removed successfully.	Pass
TC-005	Missing Value Validation	Verify missing and invalid values are handled according to defined rules.	Missing values are converted to NULL or appropriately flagged.	Missing values handled successfully during data preparation.	Pass
TC-006	Data Standardization	Verify standardization of categorical fields (e.g., patient sex, country, drug names).	Values are standardized into a consistent format.	Data standardized successfully across required fields.	Pass
TC-007	Date Validation	Verify all report dates are converted to the required DATE format.	All dates are stored in a valid DATE format.	Date conversion completed successfully.	Pass
TC-008	Dataiku Quality Check	Verify quality rules pass after data cleansing.	Dataset passes configured quality validation checks.	Quality checks completed successfully.	Pass

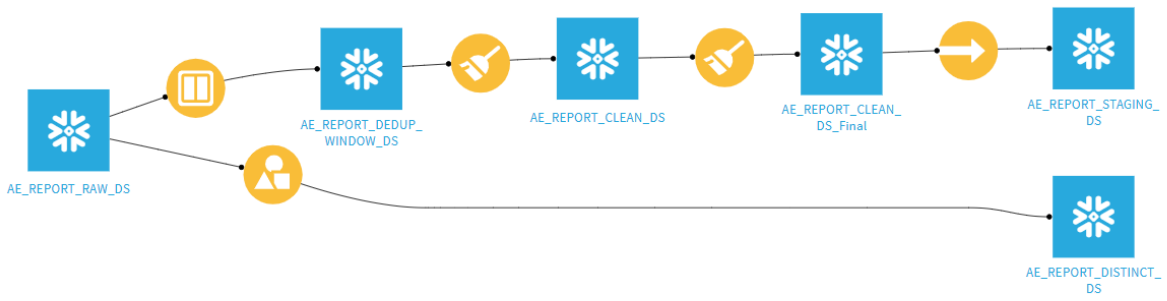


Figure 4.2 (Data Quality Testing)

4.3 Data Warehouse Testing

TC ID	Module	Test Scenario	Expected Result	Actual Result	Status
TC-009	STAGING Validation	Verify cleaned dataset is successfully loaded into the STAGING schema.	Clean dataset is available in STAGING.	Dataset loaded successfully into STAGING.	Pass
TC-010	Dimension Tables	Verify dimension tables are populated correctly.	Dimension tables contain valid and unique records.	Dimension tables populated successfully.	Pass
TC-011	Fact Table	Verify fact table contains the expected grain (one row per adverse event report).	Fact table accurately represents the defined grain.	Fact table validated successfully.	Pass
TC-012	Referential Integrity	Verify relationships between fact and dimension tables.	Foreign key relationships are consistent.	Referential integrity validated successfully.	Pass
TC-013	Analytical Views	Verify analytical views return correct aggregated results.	Views produce accurate analytical outputs.	Analytical views validated successfully.	Pass

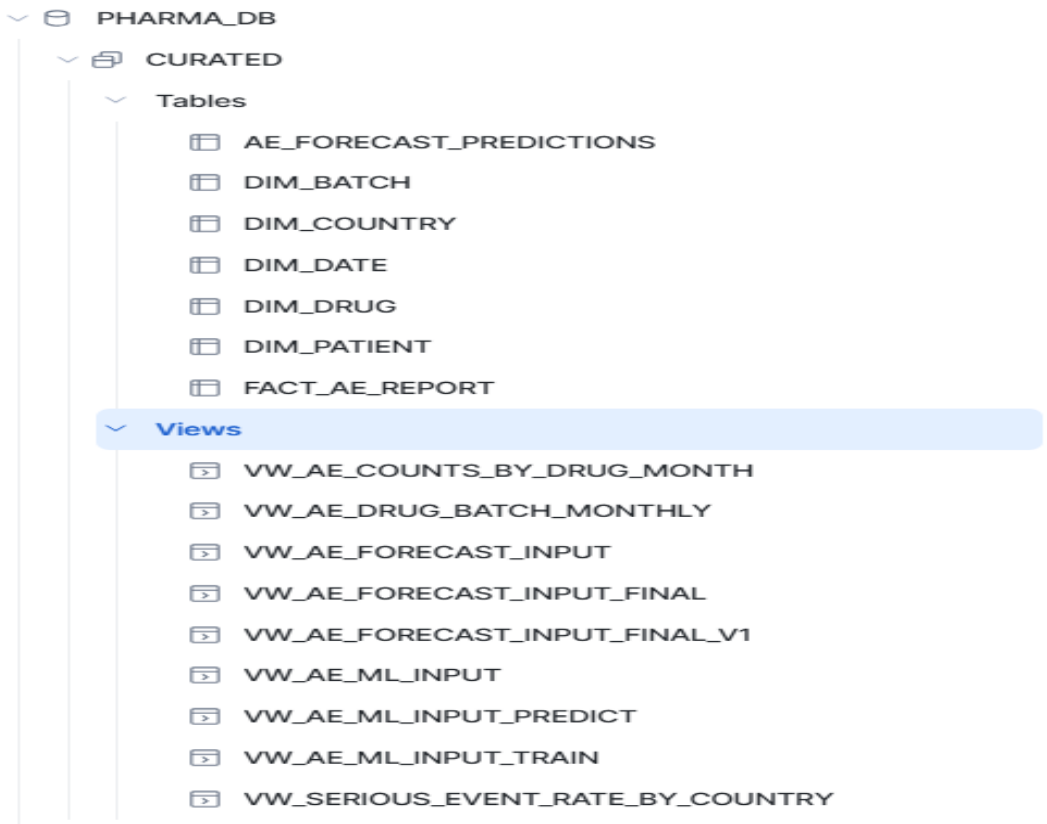
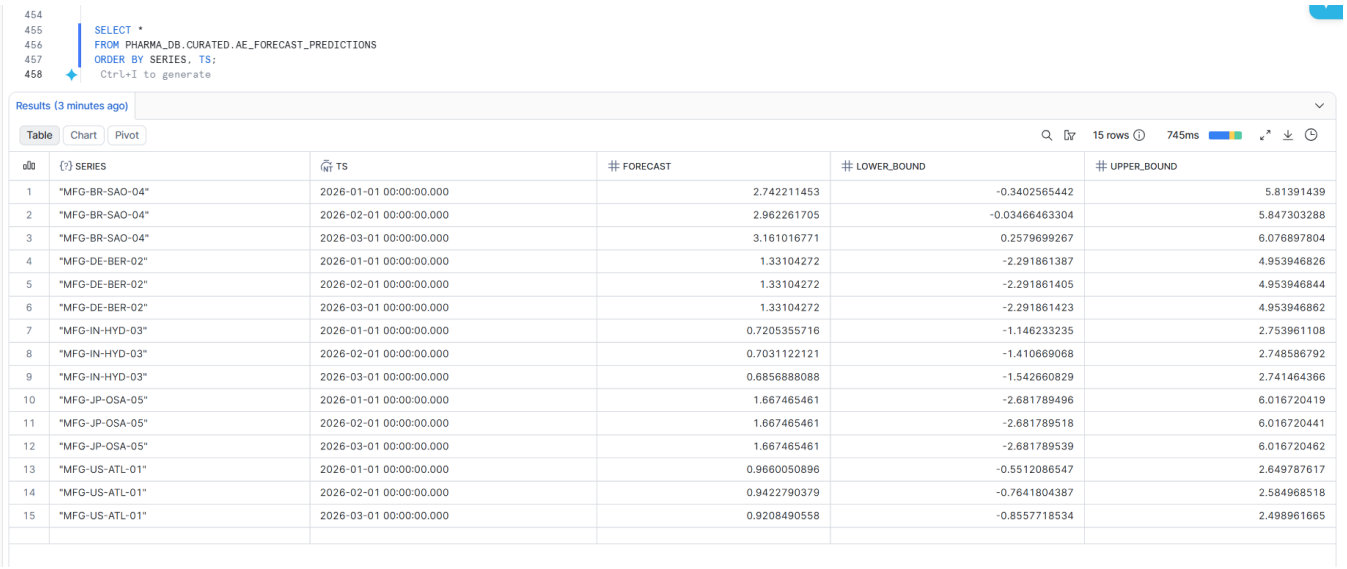


Figure 4.3: Snowflake CURATED Schema Containing Fact Table, Dimension Tables, Analytical Views, and AI/ML Views

4.4 AI/ML Validation Testing

TC ID	Module	Test Scenario	Expected Result	Actual Result	Status
TC-014	ML Input Validation	Verify forecasting input dataset is prepared correctly.	Input view contains required fields and valid data.	Forecast input dataset validated successfully.	Pass
TC-015	Cortex Forecast	Verify Snowflake Cortex Forecast model executes successfully.	Forecast model is created and executed successfully.	Forecast model executed successfully.	Pass
TC-016	Forecast Output	Verify forecast predictions are generated correctly.	Forecast results are available for the configured prediction horizon.	Prediction output generated successfully.	Pass



```
454
455   SELECT *
456   FROM PHARMA_DB.CURATED.AE_FORECAST_PREDICTIONS
457   ORDER BY SERIES, TS;
458    Ctrl+I to generate
```

Results (3 minutes ago)

Table | Chart | Pivot — 15 rows — 745ms

	SERIES	TS	FORECAST	LOWER_BOUND	UPPER_BOUND
1	"MFG-BR-SAO-04"	2026-01-01 00:00:00.000	2.742211453	-0.3402565442	5.81391439
2	"MFG-BR-SAO-04"	2026-02-01 00:00:00.000	2.962261705	-0.03466463304	5.847303288
3	"MFG-BR-SAO-04"	2026-03-01 00:00:00.000	3.161016771	0.2579699267	6.076897804
4	"MFG-DE-BER-02"	2026-01-01 00:00:00.000	1.33104272	-2.291861387	4.953946826
5	"MFG-DE-BER-02"	2026-02-01 00:00:00.000	1.33104272	-2.291861405	4.953946844
6	"MFG-DE-BER-02"	2026-03-01 00:00:00.000	1.33104272	-2.291861423	4.953946862
7	"MFG-IN-HYD-03"	2026-01-01 00:00:00.000	0.7205355716	-1.146233235	2.753961108
8	"MFG-IN-HYD-03"	2026-02-01 00:00:00.000	0.7031122121	-1.410669068	2.748586792
9	"MFG-IN-HYD-03"	2026-03-01 00:00:00.000	0.6856888088	-1.542660829	2.741464366
10	"MFG-JP-OSA-05"	2026-01-01 00:00:00.000	1.667465461	-2.681789496	6.016720419
11	"MFG-JP-OSA-05"	2026-02-01 00:00:00.000	1.667465461	-2.681789518	6.016720441
12	"MFG-JP-OSA-05"	2026-03-01 00:00:00.000	1.667465461	-2.681789539	6.016720462
13	"MFG-US-ATL-01"	2026-01-01 00:00:00.000	0.9660050896	-0.5512086547	2.649787617
14	"MFG-US-ATL-01"	2026-02-01 00:00:00.000	0.9422790379	-0.7641804387	2.584968518
15	"MFG-US-ATL-01"	2026-03-01 00:00:00.000	0.9208490558	-0.8557718534	2.498961665

Figure 4.4: Snowflake Cortex Forecast Prediction Results

4.5 Dashboard Validation

TC ID	Module	Test Scenario	Expected Result	Actual Result	Status
TC-017	Dashboard Metrics	Verify dashboard metrics match validated Snowflake data.	Dashboard values are consistent with source data.	Metrics validated successfully.	Pass
TC-018	Dashboard Visuals	Verify charts and filters display accurate information.	Visualizations render correctly and support user interaction.	Dashboard validated successfully.	Pass

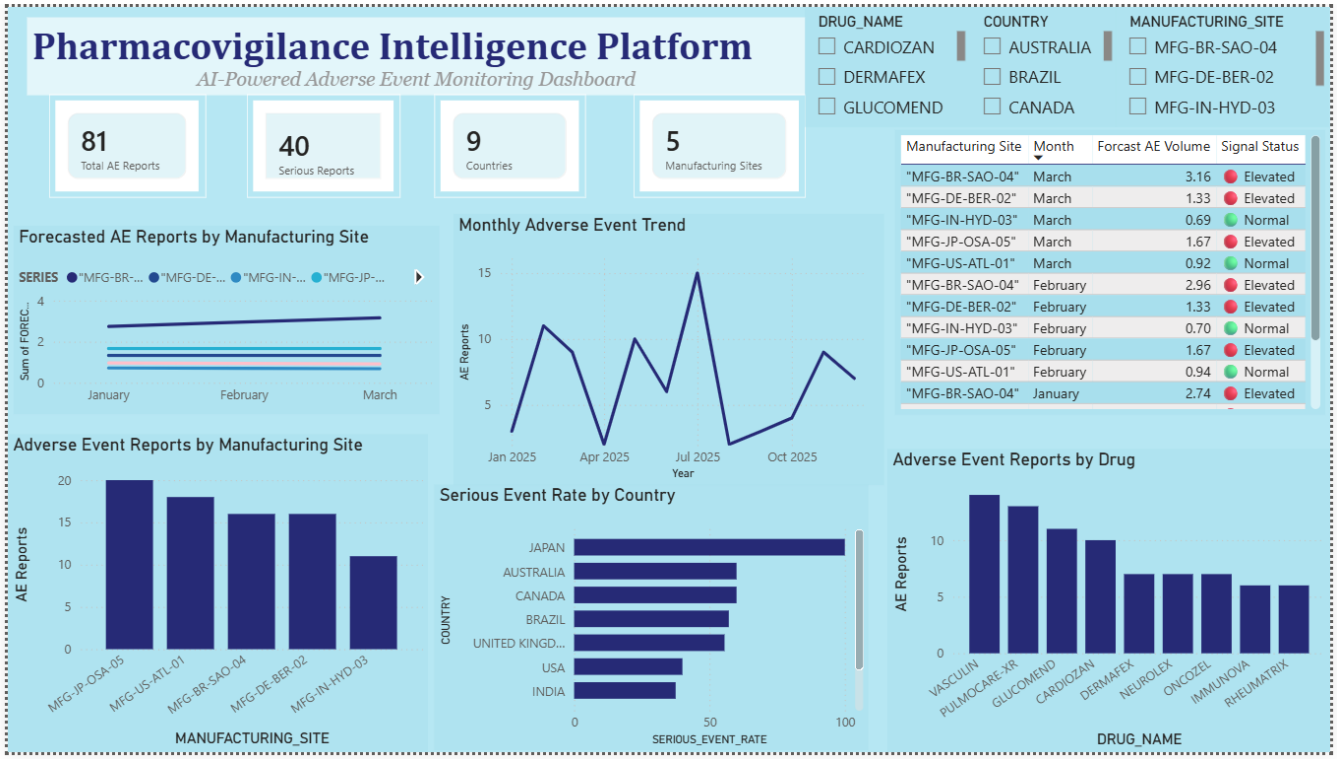


Figure 4.5: Power Bi Dashboard Analysis

5. Validation Results

The executed test cases confirmed that the Pharmacovigilance Intelligence Platform successfully met the defined functional, data quality, and analytical requirements. Validation was performed across the complete data pipeline, including data ingestion, transformation, dimensional modeling, AI/ML forecasting, and dashboard reporting.

5.1 Test Execution Summary

Metric	Result
Total Test Cases Executed	18
Test Cases Passed	18
Test Cases Failed	0
Test Cases Blocked	0
Overall Pass Rate	100%

5.2 Validation Summary

Validation Area	Validation Result	Status
Data Ingestion	Raw data was successfully loaded into the Snowflake RAW schema with matching source and target record counts.	Pass
Data Quality	Duplicate records were removed, missing values were handled, and data was standardized according to defined rules.	Pass
Data Transformation	Data cleansing and transformation logic executed successfully in Dataiku DSS.	Pass
Data Warehouse	Fact table, dimension tables, and analytical views were created and validated successfully in the CURATED schema.	Pass
AI/ML Forecasting	Snowflake Cortex Forecast executed successfully and generated forecast predictions.	Pass
Dashboard Validation	Power BI dashboard metrics and visualizations were verified against Snowflake analytical data.	Pass

5.3 Overall Validation Outcome

The testing process confirmed that all major components of the Pharmacovigilance Intelligence Platform were functioning as expected. The data pipeline successfully processed adverse event reports from ingestion through reporting, while the Snowflake warehouse, AI/ML forecasting workflow, and dashboard outputs were validated successfully.No critical defects remained open after testing, and the platform was considered ready for demonstration and analytical usage.

6. Issues Encountered & Resolution

During testing and validation, a few technical challenges were identified during data preparation, warehouse implementation, and AI/ML execution.

Issue	Resolution
Inconsistent date formats in the source dataset affected date processing.	Dates were standardized using Dataiku transformation rules.
Duplicate REPORT_ID records were identified during data quality validation.	Duplicate records were removed using Dataiku deduplication logic.
Missing and inconsistent values were present in source attributes.	Missing values were handled and categorical values were standardized during cleansing.
Temporary Snowflake Cortex execution issue occurred during model setup.	Model execution was retried after service readiness and completed successfully.

7. Conclusion

The testing and validation activities for the **Pharmacovigilance Intelligence Platform: From Raw Adverse Event Reports to AI-Powered Safety Signals** were successfully completed.

The complete data pipeline was validated, including data ingestion, data quality processing, Snowflake data warehouse components, analytical views, Snowflake Cortex Forecast workflow, and Power BI reporting.

The executed test cases confirmed that:

Raw adverse event data was successfully ingested and processed.

Data quality issues were identified and resolved through cleansing and standardization.

Snowflake fact tables, dimension tables, and analytical views were validated successfully.

AI/ML forecasting workflow executed successfully and generated prediction results.

Dashboard metrics and visualizations were consistent with validated analytical data.

All planned test cases were completed successfully with a **100% pass rate**, and no critical defects remained open after validation.

The platform is considered functionally validated and ready for demonstration and further enhancement toward production-level implementation.